Proposed Solution: Technical Deep Dive & Core Scenarios

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `3. Project Design Phase`

Phase: Solution Specification & Scenario Engineering

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1. Architectural Philosophy

NeuroClass is engineered around three non-negotiable principles:

1. **Zero-Friction Passive Operation:** The system requires no physical contact, cards, or active check-ins.
2. **Absolute Edge Privacy:** Video and audio streams never leave local host memory; raw speech is never transcribed into text.
3. **Actionable Pedagogical Assistance:** The agent moves beyond passive logging to actively support instructors during live teaching.

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2. Core Scenario 1: Sustained Presence Attendance

SCENARIO 1: SUSTAINED PRESENCE VERIFICATION FLOW	
[Incoming Detection] ---> (Is Track Duration < 15s?)	
YES	NO (Active Participant)
	v
v [Extract Face Embedding]	
[DISCARD WALK-BY]	
v	
(Match against student_db/?)	
YES	NO
	v
v [Assign Pseudonym: TRK-00X]	
[Bind STU-ID]	
+-------+------+	
v	
[Track Active Presence Time]	
v	
(Active Time / Elapsed Time >= 65%?)	
YES	NO
v v	
[STATUS: PRESENT] [STATUS: INCOMPLETE / ABSENT]	

Technical Specification (Scenario 1)

• **Transient Walk-By Filter:** Detections active for ***<15*** continuous seconds (e.g., individuals passing the open classroom door) are discarded as non-participant noise.

• **Biometric Matching & Salted Verification:** Faces are compared against pre-registered embeddings in `student_db/` using Euclidean distance thresholding (***d \le 0.55***).

• **Sustained Presence Window:** To earn verified attendance, a participant must be visible for ***\ge 65\%*** of the active session window.

• **Punctuality Enforcement:** If a student's initial verified appearance occurs ***>10*** minutes after session start, their status is flagged as **Late**.

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3. Core Scenario 2: Multimodal Engagement Analysis

SCENARIO 2: MULTIMODAL TEMPORAL FUSION PIPELINE				
MODALITY 1: VISUAL (50%)	MODALITY 2: MOTION (20%)	MODALITY 3: ACOUSTIC (30%)		
• Head Pose Yaw (	Yaw	<= 25°)	• Centroid Velocity Tracking	• Librosa RMS Energy
• Head Pose Pitch (	Pitch	<= 20°)	• Inactivity Dormancy (> 120s)	• VAD Speech Ratio
• Eye Openness / Blink Ratio	• Restless Jitter Filter (> 0.45)	• Classroom Sound Dynamics		

Fusion Mathematics

\text{Engagement Index} = (0.50 \times S_{\text{visual}}) + (0.20 \times S_{\text{motion}}) + (0.30 \times S_{\text{acoustic}})

- **Visual Attention Score (*S_{visual}*):** Proportion of active participants looking directly toward the instructional zone (***$|\text{Yaw}| \le 25^\circ \land |\text{Pitch}| \le 20^\circ$***).
- **Motion Balance Score (*S_{motion}*):** Normalizes participant bodily movement. Deductions apply when students remain completely static for ***>120***s (dormant posture) or exceed restless jitter thresholds (***>0.45***).
- **Acoustic Interaction Score (*S_{acoustic}*):** Evaluates conversational back-and-forth between instructor and learners via Voice Activity Detection (VAD). Balanced conversational dynamics maximize this component.

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4. Core Scenario 3: Context-Aware AI Teaching Suggestions

SCENARIO 3: AUTONOMOUS PEDAGOGICAL AGENT LOGIC	
[Real-Time Multimodal Telemetry Stream]	
+---> Condition A: Continuous Instructor Monologue > 12 minutes?	
	==> Trigger: "Pacing Alert: 12m monologue. Pose a quick comprehension check."
+---> Condition B: Classroom Dead Silence > 6 minutes during collaborative task?	
	==> Trigger: "Interaction Alert: 6m silence. Check if groups need facilitation."
+---> Condition C: Class-wide Engagement Drop >= 25% across 2 consecutive windows?	
==> Trigger: "Engagement Alert: Focus dropped by 28%. Suggest a 2-minute pause."	
[Cooldown Guard: Enforce strict 45-second silence between toasts to prevent alert fatigue]	

Dual-Brain Architecture (Rule-Based + Generative LLM)

1. **Deterministic Rule Engine (Primary - 100% Offline):** Instantaneous millisecond-level threshold triggers guarantee zero latency and complete offline execution.
2. **Generative LLM Engine (Augmentation - Optional Cloud):** When configured with a local or cloud LLM key (Groq LPU / LiteLLM), the agent generates enriched, pedagogical context adaptations tailored to course topics.

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5. The 3-Way Hybrid Enrollment Subsystem

To ensure zero onboarding friction in real classrooms, NeuroClass supports three distinct enrollment modalities:

3-WAY HYBRID ENROLLMENT ARCHITECTURE		
1. Static Image Upload	2. Live Webcam Snap	3. Dynamic Track Link
Instructor uploads student photo	Student stands in front of camera	Instructor views active

3-WAY HYBRID ENROLLMENT ARCHITECTURE		
via web portal. System extracts	and clicks "Capture Snap" on the	anonymous track (TRK-003)
128D embedding and saves to	UI. Instant registration creates	on live HUD and clicks
student_db/{student_id}.jpg.	student_db/{student_id}.jpg.	"[IDENTIFY / LINK] Identify / Link".

This 3-way architecture guarantees that missing pre-enrolled photos never block attendance tracking during live classes.